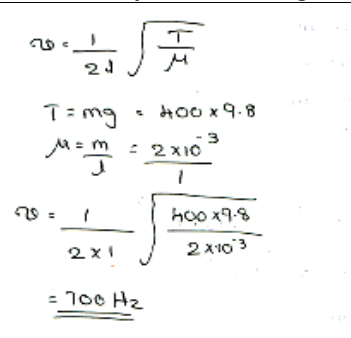


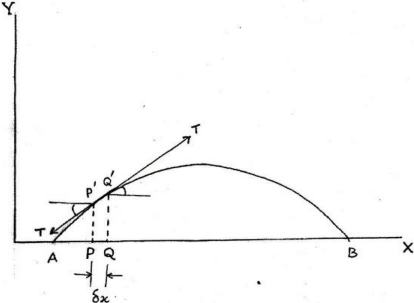
Sahrdaya College of Engineering and Technology, Kodakara		
Answer Key		
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY		
FIRST SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2021		
Course Code: PHT100		
Course Name: ENGINEERING PHYSICS A		
PART A		
		Answer all questions. Each question carries 3 marks
Q1		What is Q-factor? How is it related to angular frequency?
ans		Quality Factor (Q - factor) Q - factor is defined as 2π times the energy stored in the oscillator to the energy dissipated per cycle. ie $Q = \frac{2\pi \times \text{Energy stored in the oscillator}}{\text{Energy lost per cycle}}$ $= \frac{2\pi \times \text{Energy stored in the oscillator}}{\text{Energy lost per sec} \times \text{Time period}}$ $= \frac{2\pi E}{-\frac{dE}{dt} T}$ $Q = \frac{2\pi E}{PT} = \frac{2\pi E}{2\gamma ET} = \frac{\pi}{\gamma T} = \frac{\pi}{\gamma \frac{2\pi}{\omega}} = \frac{\omega}{2\gamma}$ $\therefore Q = \frac{\omega}{2\gamma} = \omega\tau,$ $\frac{1}{2\gamma} = \tau$, relaxation time & ω angular frequency
Q2		Calculate the frequency of the fundamental note produced by a string 1m long and weighing 2gm kept stretched by a load of 400kg.
ans		 $f = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$ $T = mg = 400 \times 9.8$ $\mu = \frac{m}{l} = \frac{2 \times 10^{-3}}{1}$ $f = \frac{1}{2 \times 1} \sqrt{\frac{400 \times 9.8}{2 \times 10^{-3}}}$ $= 700 \text{ Hz}$
Q3		Why does the central fringe of Newton's ring appear dark?

ans	At the point of contact between the lens and the glass plate, the thickness of air film $t=0$. Hence the path difference $2\mu t=0$. But at the time of reflection, the reflected wave from the glass plate undergoes a phase change of π (path difference of $\frac{\lambda}{2}$). So the waves at the centre are interfering destructively and hence a dark spot is obtained at the centre of newton's ring.
Q4	What is meant by dispersive power of grating? Give the expression with relevant terms.
ans	The dispersive power of a grating is defined as the ratio of the change in the angle of diffraction to the corresponding change in wavelength. Let two wavelengths λ and $\lambda+d\lambda$ be diffracted through angles θ and $\theta+d\theta$ respectively. Then the dispersive power of the grating is Dispersive Power, $\frac{d\theta}{d\lambda} = \frac{Nn}{\cos\theta}$ For small values of θ, $\cos\theta=1$ $\frac{d\theta}{d\lambda} = Nn$ Dispersive power is proportional to the order n and the number of lines per unit length N.
Q5	Estimate the uncertainty in the frequency of light emitted by an atom
ans	An electron exists in an excited state only for a short interval of time. Thus Δt is small, then ΔE must be large $\Delta t=10^{-8}\text{s}$ $\Delta E \cdot \Delta t = \hbar$ The uncertainty in energy, $\Delta E = \hbar/10^{-8}$ $E = h\nu$ $\Delta E = h\Delta\nu$ $\Delta\nu = \Delta E/h$ $\Delta\nu = \left[\frac{h}{10^{-8}} \right] * \frac{1}{h}$ $= 1.67 \cdot 10^7 \text{Hz}$
Q6	Describe the significance of large surface area to volume ratio of nano materials.
ans	Nano material have a relatively larger surface area to volume ratio when compared to bulk solids. Consider a sphere of radius 'r' . The surface area $= 4\pi r^2$ Its volume $= \frac{4}{3}\pi r^3$ Ratio $= \frac{\text{surface area}}{\text{volume}} = \frac{3}{r}$ Thus when the radius of the sphere decreases, its surface area to volume ratio increases. Hence as the particle size decreases a greater number of atoms are found at the surface compared to those

		inside. Thus nano material have more surface area per given volume compared with larger particles.
Q7		State Gauss' law in magnetism. Write the mathematical statement.
	ans	The surface integral of magnetic flux over a closed surface is zero. .Magnetic fields do not start or end at any point •The number of lines entering a surface equals the number of lines leaving the surface. •Magnetic monopole does not exist. $\oint \vec{B} \cdot d\vec{A} = 0$
Q8		Define divergence of a vector field. Establish its physical significance.
	ans	Multiplication of ∇ operator by another vector and the result is a scalar product. This operation is called divergence of a vector .Outflow is greater than inflow, divergence is +ve •Outflow is less than inflow, divergence is -ve •Outflow is equal to inflow, divergence is zero(in compressive fluid) $\text{div} F = \partial F_1 / \partial x + \partial F_2 / \partial y + \partial F_3 / \partial z$
Q9		What are Cooper pairs? What is their role in superconductivity?
	ans	Consider an electron that moves through the array of ions. The +ve ions gets attracted to the electrons .Hence the lattice gets distorted (e-lattice interaction). A region of increased positive charge density is produced. The region of increased positive charge density attracts another electron (lattice- e^- interaction). These 2 electrons are coupled and is called a <i>cooper pair</i>. At very low temp. these particles tend to occupy the lowest state and move together with the same velocity in an ordered form without making any collision with the lattices. Hence no resistance
Q10		What is a photo detector? Give two examples
	ans	The functions of a photo detector is to detect the optical signal . The device can be operated either in photo voltaic mode or photo conductive mode. In photoconductive mode the pn junction is connected in the reverse bias and the current is made to flow externally to complete the circuit between the junction. Commonly used photo detectors are, Photodiode PIN photodiode.
PART B		
<i>Answer any one full question from each module. Each question carries 14 marks</i>		

Module 1		
Q11	a	Derive the differential equation of a forced harmonic oscillator and find its solution. Define amplitude resonance.
ans		If an external force acts on a damped oscillatory system it is called a Forced Harmonic Oscillator. The oscillations produced, under the action of external periodic force on the body is called forced oscillations. The Equation of motion is given by $m \frac{d^2x}{dt^2} + b \frac{dx}{dt} + kx = F_0 \sin \omega_f t$ $\frac{d^2x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \frac{k}{m} x = \frac{F_0}{m} \sin \omega_f t \text{ --- (1)}$ Let $\frac{b}{m} = 2\gamma$ where γ is the damping coefft and $\omega^2 = \frac{k}{m}$ $\frac{d^2x}{dt^2} + 2\gamma \frac{dx}{dt} + \omega^2 x = f_0 \sin \omega_f t \text{ --- (2)}$ where $f_0 = \frac{F_0}{m}$ This is the differential eqn for forced harmonic oscillator $x = A \sin (\omega_f t - \theta) \text{ --- (3)}$ $\frac{dx}{dt} = A \omega_f \cos(\omega_f t - \theta)$ $\frac{d^2x}{dt^2} = -A \omega_f^2 \sin (\omega_f t - \theta)$ Putting these values in (2) $-A \omega_f^2 \sin (\omega_f t - \theta) + 2\gamma A \omega_f \cos(\omega_f t - \theta) + \omega^2 A \sin (\omega_f t - \theta) = f_0 \sin [(\omega_f t - \theta) + \theta] \text{ --- (4)}$ Expanding RHS $= f_0 \sin (\omega_f t - \theta) \cos \theta + f_0 \cos(\omega_f t - \theta) \sin \theta$ $[-A \omega_f^2 - f_0 \cos \theta + \omega^2 A] \sin (\omega_f t - \theta) + [2\gamma A \omega_f - f_0 \sin \theta] \cos(\omega_f t - \theta) = 0 \text{ --- (5)}$ To find A: For this eq. to hold good for all values of t, the coeffs of the term $\sin (\omega_f t - \theta)$ and $\cos(\omega_f t - \theta)$ must vanish separately. ie $[-A \omega_f^2 - f_0 \cos \theta + \omega^2 A] = 0$ $\therefore -A \omega_f^2 + \omega^2 A = f_0 \cos \theta \text{ --- (6)}$ $[2\gamma A \omega_f - f_0 \sin \theta] = 0$ $\therefore [2\gamma A \omega_f = f_0 \sin \theta] \text{ --- (7)}$

	<i>Squaring and eq.(6) and(7),</i> $\left(-A\omega_f^2 + \omega^2 A\right) + 4\gamma^2 A^2 \omega_f^2 = f_0^2$ $A = \frac{f_0}{\sqrt{\left(\omega^2 - \omega_f^2\right)^2 + 4\gamma^2 \omega_f^2}} \text{ --- (8)}$ <i>This is the amplitude of the Forced vibration</i> <i>Dividing eq.(7) by (6)</i> $\tan \theta = \frac{2\gamma\omega_f}{\omega^2 - \omega_f^2} \text{ --- (9)}$ <i>This gives the phase difference between the forced oscillations and the applied force.</i> <i>Substituting for A in eq.(3)</i> $x = \frac{f_0 \cdot \sin(\omega_f t - \theta)}{\sqrt{\left(\omega^2 - \omega_f^2\right)^2 + 4\gamma^2 \omega_f^2}}$ The particular frequency of the applied periodic force at which the amplitude becomes maximum is called resonant frequency. when the frequency of the applied force is equal to the natural frequency of the oscillator the amplitude will be maximum. This condition is known as resonance. Amplitude at Resonance $A_{\max} = \frac{f_0}{2\gamma\omega_f}$ The amplitude of the forced oscillations at resonance is called amplitude resonance
b	A damped oscillator of mass 1 gram has force constant 10 N/M and damping factor 1 s⁻¹. Calculate the angular frequency without damping and with damping.

ans	$k = 10 \text{ N/m}$ $m = 1 \text{ g} = 1 \times 10^{-3} \text{ kg}$ $\omega_0^2 = \frac{k}{m} = \frac{10}{10^{-3}} = 10^4$ $\omega_0 = 10^2 = \underline{100}$ $\omega^2 = \omega_0^2 - \gamma^2$ $= 100^2 - 1^2$ $= 9999, \quad \omega = \underline{99.99}$ Angular frequency without damping = 100 rad/s Angular frequency with damping = 99.99 rad/s
Q12 a	Derive an expression for fundamental frequency of transverse vibrations of a stretched string.
ans	Consider an infinitely long thin and uniform string stretched between two points by a constant tension T along the X-axis. Let the string be slightly displaced aside along the Y-axis and released. Transverse vibrations are set up in the string. Consider a small element PQ of the string of length δx. The magnitude of tension in the string will be same everywhere since the string is perfectly flexible. The tension T acts tangentially at every point on the string. $P'Q'$ is the displaced position of the string.  The tension at P' and Q' acts along the tangent making angles θ_1 and θ_2 with the horizontal. Resolving the tension along x-axis and y-axis, Then net force on PQ acting X and Y direction are $F_x = T \cos \theta_2 - T \cos \theta_1$ $F_y = T \sin \theta_2 - T \sin \theta_1$ For small oscillations θ_1 and θ_2 are very small. $\cos \theta_1 = 1 \quad ; \quad \cos \theta_2 = 1$ Also $\sin \theta_1 = \tan \theta_1$; $\sin \theta_2 = \tan \theta_2$ ($\tan \theta_1 = \sin \theta_1 / \cos \theta_1 = \sin \theta_1$) Then $F_x = 0$ $F_y = T \tan \theta_2 - T \tan \theta_1$ So the net force acting on the element δx in the displaced position is only along Y-axis. $F_y = T(\tan \theta_2 - \tan \theta_1)$ If μ is the mass per unit length of the string (linear density), mass of the element δx, $m = \mu \delta x$

$$\text{Acceleration} = \frac{\partial^2 y}{\partial t^2}$$

According to the Newton's second law of motion

$$F = ma$$

$$F = \mu \delta x \frac{\partial^2 y}{\partial t^2}$$

$$T (\tan \theta_2 - \tan \theta_1) = \mu \delta x \frac{\partial^2 y}{\partial t^2}$$

$$\text{But } \tan \theta_1 = \left(\frac{\partial y}{\partial x} \right)_x ; \quad \tan \theta_2 = \left(\frac{\partial y}{\partial x} \right)_{x+\delta x}$$

$$\text{or } \frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu} \left[\frac{\tan \theta_2 - \tan \theta_1}{\delta x} \right]$$

$$\text{ie } \frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu} \left[\frac{\left(\frac{\partial y}{\partial x} \right)_{x+\delta x} - \left(\frac{\partial y}{\partial x} \right)_x}{\delta x} \right]$$

when $\delta x \rightarrow 0$

$$\lim_{\delta x \rightarrow 0} \left[\frac{\left(\frac{\partial y}{\partial x} \right)_{x+\delta x} - \left(\frac{\partial y}{\partial x} \right)_x}{\delta x} \right] = \frac{\partial^2 y}{\partial x^2} \quad \text{--- (5)}$$

$$\frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu} \frac{\partial^2 y}{\partial x^2} \quad \text{--- (6)}$$

$$\frac{\partial^2 y}{\partial x^2} = \frac{\mu}{T} \frac{\partial^2 y}{\partial t^2}$$

This is the differential equation of a vibrating string.

Comparing this equation with the standard wave equation

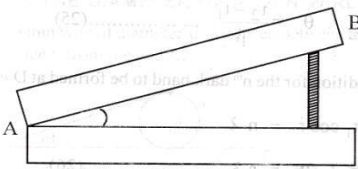
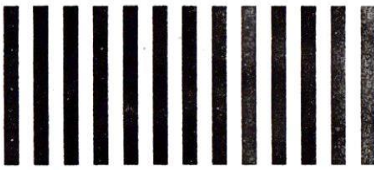
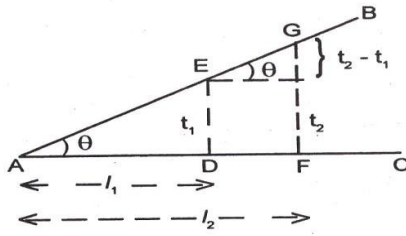
$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$$

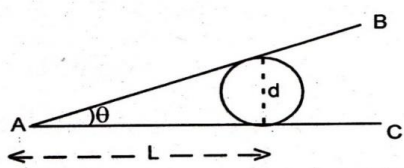
$$\begin{aligned} \frac{1}{v^2} &= \frac{\mu}{T} \\ v^2 &= \frac{T}{\mu} \\ v &= \sqrt{\frac{T}{\mu}} \end{aligned}$$

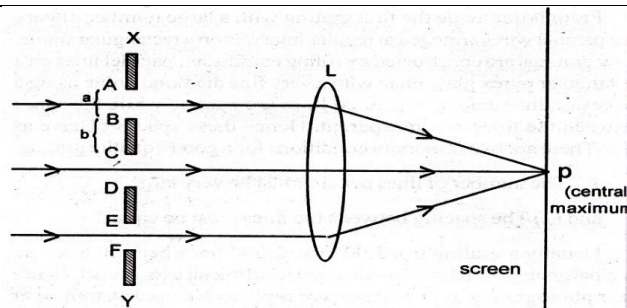
This is the velocity of transverse wave on a stretched string.

$$v = v\lambda$$

		$v = \frac{v}{\lambda}$ $v = \frac{1}{\lambda} \sqrt{\frac{T}{\mu}}$ This is the frequency of transverse waves developed on a stretched string. Modes of vibrations If the string is 'l' meters long and vibrating in one segment only, then $l = \frac{\lambda}{2}$  $\lambda = 2l$ The frequency of transverse wave is given by $v = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$ Frequency of the standing wave with n antinodes is given by $v = \frac{n}{2l} \sqrt{\frac{T}{\mu}}$ Where , n is the no. of antinode <table> <tr> <td>n = 1</td><td>1st Harmonic or fundamental frequency</td></tr> <tr> <td>n = 2</td><td>2nd Harmonic</td></tr> <tr> <td>n = 3</td><td>3rd Harmonic</td></tr> </table>	n = 1	1 st Harmonic or fundamental frequency	n = 2	2 nd Harmonic	n = 3	3 rd Harmonic
n = 1	1 st Harmonic or fundamental frequency							
n = 2	2 nd Harmonic							
n = 3	3 rd Harmonic							
b		A wave is represented by $y = 3 \times 10^{-3} \cos (8.4 \times 10^{13} t + 2.8 \times 10^5 z)$ where y and z are in m and t in second. Compute the following (i) amplitude (ii) frequency (ii) wavelength (iv) wave velocity						
ans		$y = 3 \times 10^{-3} \cos (8.4 \times 10^{13} t + 2.8 \times 10^5 z)$ $y = a \cos (kz + \omega t)$ $a = 3 \times 10^{-3}, \omega = 8.4 \times 10^{13}, k = 2.8 \times 10^5$ $\omega = 2\pi \nu$ $\nu = \frac{\omega}{2\pi} = \frac{8.4 \times 10^{13}}{2\pi}$ $\nu = 1.33 \times 10^{13} \text{ Hz}$ $a = 3 \times 10^{-3} \text{ m}$ $\lambda = \frac{2\pi}{k} = \frac{2\pi}{2.8 \times 10^5} = 2.24 \times 10^{-5} \text{ m}$ $v = \frac{\omega}{k} = \frac{8.4 \times 10^{13}}{2.8 \times 10^5} = 3 \times 10^8 \text{ m/s}$						
Module 2								
Q13		Explain the formation of interference fringes in air wedge. How is it used to determine the diameter of a thin wire?						

a	
ans	INTERFERENCE IN A WEDGE SHAPED FILM  Two plane glass plates are placed such that they are in contact at one end and separated by a small distance at the other end. A wedge shaped air film is formed between them. A beam of monochromatic light is incident normally on the glass plate. A part of the light is reflected from the top surface of the thin film another part of light is reflected from the top surface of the lower glass plate. These two reflected beams interfere. A system of equidistant, parallel, dark and bright bands are observed. Since thickness of air film is same along a line parallel to the line of contact of the glass plate, the interference pattern are parallel and straight to the line of contact. Each fringe is the locus of all points where the thickness of air film has a constant value. The angle between the glass plates is called ANGLE OF WEDGE (θ in radian).   Let nth dark band is formed at D where thickness of air film is t_1 and $(n+1)$th dark band be formed at F where thickness of film is t_2. Let $AD=l_1$ and $AF=l_2$ $\tan \theta = t_1 / l_1 = t_2 / l_2$ $\tan \theta = (t_2 - t_1) / (l_2 - l_1)$ Since θ is very small, $\tan \theta = \theta$ $\theta = (t_2 - t_1) / (l_2 - l_1)$ But $l_2 - l_1 = \beta$, the band width, the distance between two successive dark bands. $\theta = \frac{t_2 - t_1}{\beta}$ Condition for the nth dark band to be formed at D $2\mu t_1 \cos r = n\lambda$ $\mu = 1 \quad \& \cos r = 1$ $2t_1 = n\lambda \dots\dots(1)$ For $(n+1)$th dark band at F $2t_2 = (n+1)\lambda \dots\dots(2)$

		Equation (2) -(1) $2(t_2 - t_1) = \lambda$ $(t_2 - t_1) = \lambda/2$ Then $\theta = \frac{\lambda}{2\beta}$ If the medium between the glass plate has refractive index μ, $\theta = \frac{\lambda}{2\mu\beta}$ DIAMETER OF THE THIN WIRE  Let a thin wire of diameter d be placed between the glass plates at a distance L from the end A $\theta = d/L \text{ and } \theta = \lambda/2\beta$ $d/L = \lambda/2\beta$ $d = \frac{\lambda L}{2\beta}$
	b	The diameter of the 10th and 20th Newton's rings formed with a plano - convex lens and an optically plane glass plate are $0.415 \times 10^{-2} \text{m}$ and $0.616 \times 10^{-2} \text{m}$ respectively. If the wavelength of the interfering light is $5893 \text{\AA}$, calculate the radius of curvature of the lens.
	ans	$D_{10} = 0.415 \times 10^{-2} \text{m}$ $D_{20} = 0.616 \times 10^{-2} \text{m}$ $\lambda = 5893 \times 10^{-10} \text{m}$ $R = \frac{D_{n+k}^2 - D_n^2}{4n\lambda}$ $= \frac{(0.616 \times 10^{-2})^2 - (0.415 \times 10^{-2})^2}{4 \times 10 \times 5893 \times 10^{-10}}$ $= 0.8579 \text{m}$
	Q14 a	Discuss the theory of plane transmission grating and also derive the grating equation. State and explain Rayleigh's criterion for the limit of resolution in the case of grating.
	ans	PLANE TRANSMISSION GRATING is a plane glass plate containing a large number of equidistant parallel lines drawn using a fine diamond point . The space between the lines acts as narrow slits through which light is transmitted. The lines are opaque to light. Grating is an arrangement of a large number of parallel slits of equal width separated by equal opaque spaces . Usually a grating has 5000 to 12000 lines per cm. There are two important conditions for a good quality grating. 1. The number of lines per cm must be very large. 2. The spacing between the lines must be equal.



Consider a plane transmission grating placed perpendicular to the plane of paper. AB represents a slit and BC represents a line. Let 'a' be the width of each slit and 'b' be the width of each line. The distant (a+b) is called grating element or grating constant.

A plane wavefront of monochromatic light of wavelength λ is falling normally on this slit. Each point of the wave front sends out secondary waves in all direction. The straight and parallel waves from each point can be focussed on the screen using lens. These straight waves path difference will be zero. They will interfere constructively producing brightness at the centre. This central bright band is called central maximum.

The position of central maximum is same for all the wavelength. The central maxima will have the same colour as the incident light.

Consider two waves diffracted from two points A and C of slit. They travel along AM and CN. Draw CK perpendicular to AM. Then the path difference between the two waves is AK.

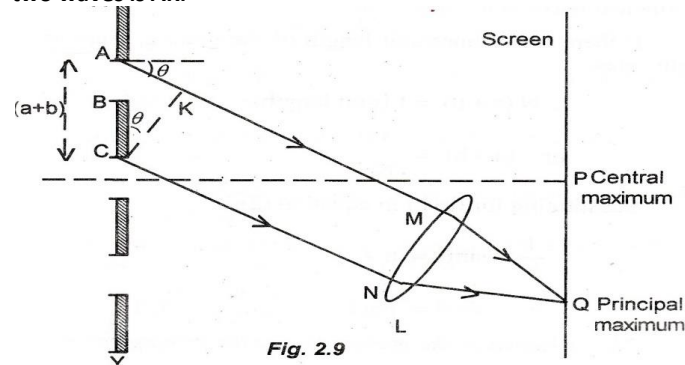


Fig. 2.9

From triangle ACK

$$\sin \theta = AK/AC$$

$$AK = AC \sin \theta$$

$$AK = (a+b) \sin \theta$$

$$\text{If } (a+b) \sin \theta = n\lambda \dots\dots(1)$$

where $n=0,1,2,3,\dots$ two waves interfere constructively. This is called principle maximum. For different values of n , there are different values of θ .

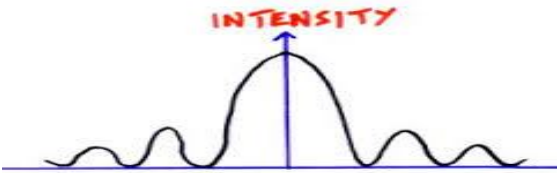
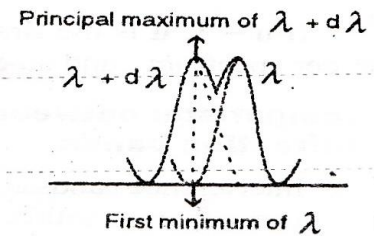
If $n=1$, it is the first order principle maximum, if $n=2$, it is the second order principal maximum and so on. Thus on either side of central maximum, a number of principal maxima are obtained.

If there are N lines/unit length of the grating, there are N slit also.

$$N(a+b) = 1 \text{ (unit length)}$$

$$(a+b) = 1/N$$

Substitute in equation (1)

		$\frac{1}{N} \sin \theta = n\lambda$ $\sin \theta = Nn\lambda$ This is known as grating law or grating equation.  For a grating θ is different for different colours (λ), for each value of n. If white light is used, it gets split up into different colours. According to Rayleigh's criterion for resolution, two neighbouring spectral lines will be just resolved when the principal maximum of one in any order falls on the first minimum of the other in the same order.  Let λ and $\lambda + d\lambda$ be the wavelengths corresponding to two neighbouring spectral lines of the same order. Then, the two spectral lines are visible as separate when the principal maximum of wavelength $\lambda + d\lambda$ falls on the first minimum of wavelength λ of the same order.
	b	Light of wavelength 656nm falls normally on a grating 20mm wide. The first order is 18° from the normal. What is the total number of lines in the grating?
	ans	$\lambda = 656 \times 10^{-9} \text{ m}, \theta = 18^\circ$ $d = 20 \times 10^{-3} \text{ m}, n = 1$ $\sin \theta = Nn\lambda$ $N = \frac{\sin \theta}{n\lambda} = \frac{\sin(18^\circ)}{1 \times 656 \times 10^{-9}}$ $N = 4.710 \times 10^5 \text{ lines/m}$ $d = 20 \times 10^{-3} \text{ m}$ Total number of lines $N_1 = 20 \times 10^{-3} \times 4.7106 \times 10^5$ $= 9421.2 \text{ lines}$
Module 3		ans
	Q15 a	Derive the Schrodinger's time dependent equation for a moving particle and hence derive the Schrodinger's time independent equation
	ans	Schrodinger's equation is the basic expression in quantum mechanics. We can derive it by considering the plane wave equation and combining with Einstein's eqn for quantum energy and

de Broglie expression for wavelength.

A particle in motion is associated with a wave function that contains the information about the motion.

A plane wave that propagates along x- direction is given by

$$\Psi = Ae^{i(kx - \omega t)}$$

where k is the wave vector, $k = 2\pi/\lambda$

ω is the angular frequency, $\omega = 2\pi\nu$

Einstein's formula for photon energy is

$$E = h\nu = \frac{h\omega}{2\pi} = \hbar\omega; \text{ where } \hbar = \frac{h}{2\pi}$$

de-Broglie's expression for matter wavelength is $\lambda = \frac{h}{p}$

$$\therefore p = \frac{h}{\lambda} = \frac{h}{2\pi} \cdot \frac{2\pi}{\lambda} = \hbar k$$

Using the above expression, equation (1) becomes

$$\Psi = Ae^{\frac{i}{\hbar}(\hbar kx - \hbar\omega t)}$$

$$\text{ie } \Psi = Ae^{\frac{i}{\hbar}(px - Et)}$$

On partial differentiation of Ψ with respect to x, twice, we get

$$\frac{\partial^2 \Psi}{\partial x^2} = \frac{-p^2}{\hbar^2} \Psi \dots \dots \dots (2) \quad \text{and}$$

Differentiating with respect to time,

$$\frac{\partial \Psi}{\partial t} = \frac{-iE}{\hbar} \Psi \dots \dots \dots (3)$$

$$p^2 \Psi = -\hbar^2 \frac{\partial^2 \Psi}{\partial x^2} \dots\dots\dots(4)$$

$$= -i\hbar \frac{\partial}{\partial x} \left(-i\hbar \frac{\partial}{\partial x} \right) \Psi \dots\dots\dots(4a)$$

From 3 and 4, $p = -i\hbar \frac{\partial}{\partial x}$

This is called momentum operator:

$$E \Psi = i\hbar \frac{\partial \Psi}{\partial t} \dots\dots\dots(5)$$

$$\therefore E = i\hbar \frac{\partial}{\partial t} \dots\dots\dots(5a)$$

This is called Energy operator.

For a free particle total energy is given by

$$E = \frac{p^2}{2m}, \text{ since } V = 0$$

$$\text{ie } E \Psi = \frac{p^2}{2m} \Psi \dots\dots\dots(6)$$

Using equation (4) and (5), Equation (6) becomes

$$\frac{-\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} = i\hbar \frac{\partial \Psi}{\partial t} \dots\dots\dots(7)$$

This is Schrodinger's equation for a free particle in one dimension.

In three dimension it becomes for a free particle,

$$\frac{-\hbar^2}{2m} \nabla^2 \Psi = i\hbar \frac{\partial \Psi}{\partial t} \dots\dots\dots(8)$$

$$\text{where } \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

is called *Laplacian operator*

If the particle is moving under a potential $V(r,t)$ then equation (8) becomes

$$\left[\frac{-\hbar^2}{2m} \nabla^2 + V_{(r,t)} \right] \Psi = i\hbar \frac{\partial \Psi}{\partial t}$$

This is Schrodinger's time dependent equation

TIME INDEPENDENT SCHRÖDINGER EQUATION (STEADY STATE EQUATION)

According to Schrodinger, de- Broglie's wavelength holds good for any particle moving in any field of force with potential energy V .

Then total energy $E = \text{Kinetic energy} + \text{Potential Energy}$

$$E = \frac{1}{2} m v^2 + V$$

$$= \frac{p^2}{2m} + V$$

$$\text{ie } p^2 = 2m(E - V)$$

$$\text{or } p = [2m(E - V)]^{1/2}$$

The wave equation in Cartesian co-ordinate system can be written as

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} = \frac{1}{u^2} \frac{\partial^2 \Psi}{\partial t^2} \dots\dots\dots(9)$$

$$\text{where } \Psi = \Psi_0 e^{-i\omega t} \dots\dots\dots(10)$$

'u' is the velocity of motion of wave and $\Psi(x, y, z, t)$ represents the amplitude of the wave associated with the particle

From (10), we get

$$\frac{\partial^2 \Psi}{\partial t^2} = -\omega^2 \Psi$$

So Equation (9) becomes

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} = \frac{-\omega^2}{u^2} \Psi \dots\dots\dots(11)$$

$$\frac{\omega^2}{u^2} = \frac{(2\pi\nu)^2}{(\nu\lambda)^2} = \frac{4\pi^2}{\lambda^2}$$

So we have

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} = \frac{-4\pi^2}{\lambda^2} \Psi \quad \text{or}$$

$$\nabla^2 \Psi + \frac{4\pi^2}{\lambda^2} \Psi = 0 \dots\dots\dots(12)$$

Equation (12) is a general equation which is independent of time.

Let us now introduce the concept of de-Broglie wavelength.

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m(E - V)}}$$

$$\lambda^2 = \frac{h^2}{2m(E - V)}$$

Substituting the wavelength in eqn (12)

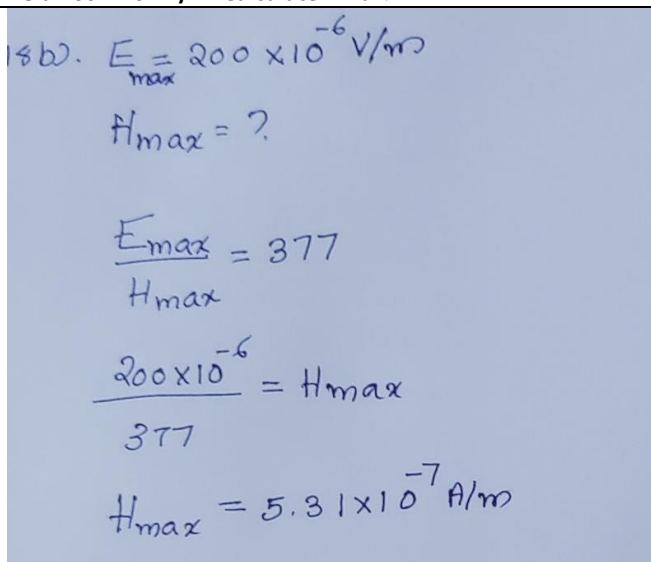
		$\nabla^2 \Psi + \frac{4\pi^2}{h^2} [2m(E - V)] \Psi = 0$ or $\nabla^2 \Psi + \frac{8\pi^2 m}{h^2} (E - V) \Psi = 0$ or $\nabla^2 \Psi + \frac{2m}{\hbar^2} (E - V) \Psi = 0 \dots\dots\dots (13)$ <i>This represents Schrodinger's time - independent wave equation.</i> For the case of a free particle (V=0), the Schrodinger equation becomes $\nabla^2 \Psi + \frac{2mE}{\hbar^2} \Psi = 0$
	b	If an electron's position can be measured to an accuracy of 2.0×10^{-8} m, how accurately can its velocity be known?
	ans	$\Delta x \Delta p_x = \hbar$ $\Delta x \cdot m \Delta v = \hbar$ $\Delta v = \frac{\hbar}{m \Delta x}$ $= \frac{h}{2\pi \times m \times \Delta x}$ $= \frac{6.626 \times 10^{-34}}{2 \times 3.14 \times 9.1 \times 10^{-31} \times 2 \times 10^{-8}}$ $= 0.0579 \times 10^5 \text{ m/s}$ $= \underline{\underline{5.79 \text{ km/s}}}$
Q16	a	Give a brief note on mechanical, optical and electrical properties of nanomaterials. Mention any four applications of nanotechnology
	ans	Electrical properties • Ionization potential at nano sizes are higher than that at bulk solids. • The change in electrical properties is not the same in all nano materials. • In nano ceramics the electrical conductivity increases with the decrease in particle size. • But in metals it decreases. • In the case of bulk metals the energy level of electron is continuous whereas in nano sized metals the energy levels are discrete. Optical properties • Depending on the particle size, same substance shows different colours. • Gold spheres of 100 nm size appear orange while that of 50 nm size appears green The particles can be made to emit or absorb specific wavelength of light by varying its size Mechanical properties • Nano crystalline nickel is as strong as hardened steel. Because of nano size many

		mechanical properties such as hardness, elastic modulus, fracture toughness, scratch resistance and fatigue strength are modified • 30-50 % lower elastic moduli than conventional materials • 2-7 times higher hardness • Super plastic behaviour in brittle ceramics Applications of Nanotechnology <u>Nanomedicine</u> • Cell imaging • Treating cancer • For diagnostic purposes • The drug consumption and side effects can be reduced significantly by targeted medicine. <u>In Energy</u> 1. By reducing energy consumption by designing LEDs or Quantum Caged Atoms(QCAs) etc. 2. By increasing energy production efficiency by designing specific catalysts to improve the combustion efficiency in IC engines <u>Information and communication</u> Memory storage devices such as Nano-RAM is developed using nanotubes <u>Textiles</u> • Clothes made of nanofibers are stain-repellent and wrinkle-free • These textiles can be washed less frequently
	b	Explain Quantum confinement.
	ans	Quantum confinement is the restricted motion of randomly moving electron in specific energy levels when the dimensions of a material approach the de Broglie wavelength of electron. When the material size is sufficiently small(10nm or less) the electrons and holes are squeezed into a dimension that approaches a quantum measurement called "Exciton Bohr Radius". When the particle size approaches the Bohr exciton radius, the quantum confinement effect increases. Based on number of dimensions that are confined, nano structure is classified in to 4 categories 1. Zero dimensional (0D) nano structured material:- If all the three dimensions become nanometer range, we obtain a Zero dimensional nano structure called nano particles 2. One dimensional (1D) nano structured material:- If two dimensions are restricted and one dimension remains large, then the resulting structure is one dimensional nano structure called nanowires, nano rods and nano tube 3. Two dimensional (2D) nano structured material:- If one dimension is reduced to the nano range while the other two dimensions remain large, then a two dimensional nano structured material are obtained (nano sheet) 4. Three dimensional (3D) nano material:- all the 3 dimensions become large (bundles of nano wires)
Module 4		
	Q17 a	Define the terms magnetisation, magnetic flux density, magnetic permeability, relative permeability and susceptibility. Obtain the relation between relative permeability and susceptibility.

ans	MAGNETIC FLUX DENSITY (B) - The total Number of magnetic field lines passing perpendicularly through unit area. $B = \frac{\phi}{A}$ $\phi = BA$ - Expressed Weber/sq.meter Or Tesla MAGNETISATION (M) - Measure of the magnetisation of a magnetised specimen. - Defined as the magnetic moment per unit volume of the material. $M = m/v$ MAGNETIC SUSCEPTIBILITY - Indicates degree of magnetization of a material. - Ratio of magnetization M to the applied magnetizing field intensity H. - Dimensionless quantity. $M = \chi H$ MAGNETIC PERMEABILITY - Defined as the property of the material to allow the magnetic lines of force to pass through it. - Ratio of the flux density B to the field intensity H. $\mu = \frac{B}{H}$ $B = \mu_0(H + M)$ $B = \mu_0 H + \mu_0 M$ $\frac{B}{H} = \frac{\mu_0 H}{H} + \frac{\mu_0 M}{H}$ $\mu = \mu_0 + \mu_0 \chi$ $\mu = \mu_0(1 + \chi)$ $\frac{\mu}{\mu_0} = (1 + \chi)$ $\mu_r = 1 + \chi$
b	Calculate the magnetic flux density and the magnetic moment per unit volume when a magnetising field of 6×10^5 A/m applied. Magnetic susceptibility is -8.2×10^{-6}.

ans	$17b). H = 6 \times 10^5 \text{ A/m.}$ $\chi = -8.2 \times 10^{-6}$ $\chi = \frac{M}{H}$ $M = \chi H = -8.2 \times 10^{-6} \times 6 \times 10^5 = -4.92 \text{ A/m.}$ $B = (1 + \chi) \mu_0 H = (1 - 8.2 \times 10^{-6}) \mu_0 \times 6 \times 10^5$ $= 0.75 \text{ Wb/m}^2.$
Q18 a	Derive Maxwell's equations from the fundamental laws of electricity and magnetism.
ans	MAXWELL EQUATIONS $\nabla \cdot D = \rho$ $\nabla \cdot B = 0$ $\nabla \times E = -\frac{\partial B}{\partial t}$ $\nabla \times H = J + \frac{\partial D}{\partial t}$ <u>Derivation of 1st equation</u> According to Gauss's law in electrostatics, The total normal electric flux through a closed surface is equal to $\frac{1}{\epsilon_0}$ times the total charge enclosed by the surface. That is $\phi_E = \frac{q}{\epsilon_0}$ $\int E \cdot ds = \frac{q}{\epsilon_0}$ $\int E \cdot ds = \frac{1}{\epsilon_0} \int \rho dv$ $\int \epsilon_0 E \cdot ds = \int \rho dv$ $\epsilon_0 E = D$ is called electric displacement vector $\int D \cdot ds = \int \rho dv$ Applying Gauss theorem $\int \nabla \cdot D dv = \int \rho dv$ To hold this equation the integrands must be equal. $\nabla \cdot D = \rho$.....Maxwell's First equation <u>Maxwell's Second Equation</u> By Gauss's law in magnetism the net magnetic flux emerging through any closed surface is zero. Then $\int B \cdot ds = 0$ By using Gauss theorem, $\int \nabla \cdot B dv = 0$ $\nabla \cdot B = 0$ For a magnetic dipole, in any closed surface the magnetic flux directed inwards towards South

	Pole will be equal to the flux outward from North Pole. The net flux will be zero for a dipole. Maxwell's Third Equation According to Faradays law of electromagnetic Induction, Induced emf $e = -\frac{\partial \phi}{\partial t}$ ϕ is magnetic flux $e = -\frac{\partial}{\partial t} \int B \cdot ds \dots\dots\dots(1)$ We know that $e = \int E \cdot dl \dots\dots\dots(2)$ $\int E \cdot dl = -\frac{\partial}{\partial t} \int B \cdot ds$ Using Stokes theorm $\int \nabla \times E \cdot ds = -\frac{\partial}{\partial t} \int B \cdot ds$ To hold this equation Integrands must be equal $\nabla \times E = -\frac{\partial B}{\partial t} \dots\dots\dots \text{Maxwell's third equation}$ Maxwell's Fourth Equation According to amperes circuital law the magnetic induction around a closed current carrying loop is equal to μ_0 times the current flowing in the loop. Thus $\int B \cdot dl = \mu_0 I$ $\int \mu_0 H \cdot dl = \mu_0 I$ $\int H \cdot dl = I \dots\dots\dots(1)$ If J is the current density of the loop, then current $I = \int J \cdot ds \dots\dots\dots(2)$ From eqn 1 and 2 $\int H \cdot dl = \int J \cdot ds$ Using Stokes theorm $\int \nabla \times H \cdot ds = \int J \cdot ds$ $\nabla \times H = J \dots\dots\dots(4)$ Take divergent of eqn (4) $\nabla \cdot \nabla \times H = \nabla \cdot J$ But $\nabla \cdot \nabla \times H = 0$ Since divergent of curl of any vector will be zero. Hence $\nabla \cdot J = 0 \dots\dots\dots(5)$ From eqn of continuity $\nabla \cdot J = -\frac{\partial \rho}{\partial t} \dots\dots\dots(6)$ Here there is incompatibility in eqn 5 and 6 Hence Maxwell introduced a new term J_D called displacement current density in eqn no (4) $\nabla \times H = J + J_D \dots\dots\dots(7)$ Now take divergent on both sides of eqn (7) $\nabla \cdot \nabla \times H = \nabla \cdot J + \nabla \cdot J_D$ $0 = \nabla \cdot J + \nabla \cdot J_D$ $-\nabla \cdot J_D = \nabla \cdot J$ $\nabla \cdot J = -\nabla \cdot J_D$ From eqn 6
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		$\frac{\partial \rho}{\partial t} = -\nabla \cdot \mathbf{J}_D$ From Maxwell's first law $\rho = \nabla \cdot \mathbf{D}$ $\frac{\partial}{\partial t} \nabla \cdot \mathbf{D} = \nabla \cdot \mathbf{J}_D$ $\nabla \cdot \frac{\partial \mathbf{D}}{\partial t} = \nabla \cdot \mathbf{J}_D$ $\frac{\partial \mathbf{D}}{\partial t} = \mathbf{J}_D$ now substitute this value in eqn 7 $\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \dots \dots \dots \text{This is Maxwell's fourth equation}$
	b	Plane electromagnetic wave (sinusoidal) has maximum intensity of electric field $200 \times 10^{-6} \text{ V/m}$. Calculate $H_{\max}$.
	ans	 18b). $E_{\max} = 200 \times 10^{-6} \text{ V/m}$ $H_{\max} = ?$ $\frac{E_{\max}}{H_{\max}} = 377$ $\frac{200 \times 10^{-6}}{377} = H_{\max}$ $H_{\max} = 5.31 \times 10^{-7} \text{ A/m}$
Module 5		
	Q19 a	Describe the phenomenon of superconductivity. Define critical temperature and critical magnetic field. Mention any four applications of superconductors
	ans	The property by which the resistivity of a material suddenly falls to zero when they are cooled below a particular temperature is called superconductivity. <u>CRITICAL TEMPERATURE (T_c)</u> It is that temperature at which resistivity of a superconducting material suddenly falls to zero and becomes a superconductor. it is the temperature at which a material changes from normal state to superconducting state When the temperature is increased above T_c, it attains resistivity and attains its normal state. When the temperature is decreased below T_c, it attains zero resistivity and attains its superconducting state <u>CRITICAL FIELD (H_c)</u> The minimum strength of magnetic field to destroy superconducting nature. <u>EFFECT OF MAGNETIC FIELD- CRITICAL FIELD (H_c)</u>

$$H_c = H_0 \left[1 - \left(\frac{T}{T_c} \right)^2 \right]$$

MEDICAL FIELD

- .In MRI machine
- .to remove tumour cells
- .diagnosis of epilepsy

ELECTRIC ENERGY AND POWER TRANSMISSION

- .electric power can be transmitted through superconducting cables without transmission loss
- .electrical motors
- .electrical generators

COMPUTER FIELD

- superconducting wires
 - storage devices in computer
- super computer

ELECTRONICS AND SMALL DEVICES

SQUID

JOSEPHSON EFFECT

b Distinguish between Type I and Type II superconductors.

ans

$$B = \mu_0(H + M)$$

When $B=0$
 $H = -M$

TYPE 1 AND TYPE 2 SUPER CONDUCTORS

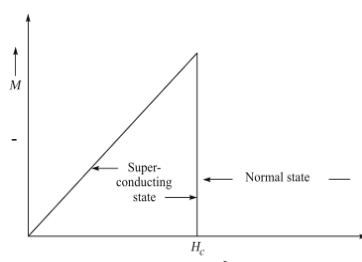


Fig. 19.5 Type I superconductor

- ☐ Upto H_c repulsive force(-M) present.
- ☐ At H_c repulsive force suddenly zero.
- ☐ After that ($H > H_c$) magnetic field start to penetrates

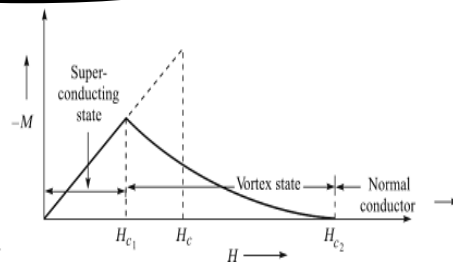


Fig. 19.6 Type II superconductor

- ☐ From H_{c1} to H_{c2} (vortex state) repulsive force(-M) start to decrease slowly and at H_{c2} only it becomes zero and after that penetration of magnetic field occurs

		TYPE - I • SUDDEN CHANGE: TRANSITION TO NORMAL STATE IS VERY ABRUPT • WHEN MAGNETIC FIELD IS REDUCED BELOW H_c THE MATERIAL ACQUIRES SUPERCONDUCTING PROPERTY • H_c HAS VERY LOW VALUE (0.1 Tesla) • CALLED AS SOFT SUPERCONDUCTORS BECAUSE SUPERCONDUCTIVITY CAN BE EASILY DESTROYED • E.g.: Hg, Pb, Al, Mg, Tin, AND TYPE - II • GRADUAL CHANGE: TRANSITION TO NORMAL STATE IS GRADUAL • H_{c2} HAS VERY LARGE VALUE (50 Tesla) • CALLED AS HARD SUPERCONDUCTORS REQUIRE HIGH MAGNETIC FIELD TO REGAIN NORMAL STATE • USED TO PRODUCE VERY HIGH MAGNETIC FIELD >30 Tesla
Q20	a	Define acceptance angle and numerical aperture of optic fibre. Derive an expression for numerical aperture of a step index fibre.
ans		ACCEPTANCE ANGLE: (θ_a) Acceptance angle (θ_a) is the maximum angle at the edge of the fibre within which if the light enters the fibre it will be propagated by total internal reflection. NUMERICAL APERTURE N.A is the light gathering power of optic fibre . It measures the amount of light accepted by the fibre $N.A = \sin \theta_a \quad NA = \sqrt{n_1^2 - n_2^2}$ Consider a light ray AB incident at B at the edge of the core of an optic fibre from air. It is incident at an angle θ_a with the axis of the core. This maximum angle is known as acceptance angle. Since it travels from air to core, it is refracted along BC at an angle θ_r. This refracted ray is now incident at C at the core-cladding interface with an angle slightly greater than the critical angle θ_c. Hence the ray is undergoing total internal reflection and it is travelling along CD. Thus the ray is propagated through the fibre by multiple total internal reflections. At C , CN is drawn normal to the axis. The angle at C is taken as the limiting angle θ_c , the critical angle. Let n_0 be the refractive index of air, n_1 that of core and n_2 that of cladding. By Snell's law at B, $\sin \theta_a \sin \theta_r = n_1 n_0$ $n_0 \sin \theta_a = n_1 \sin \theta_r$ n_0 for air = 1 and $NA = \sin \theta_a$ $NA = n_1 \sin \theta_r$(1) At the critical angle, $\sin \theta_c / \sin 90 = n_2 / n_1$

		$\sin \theta_r = \sqrt{\frac{n_1^2 - n_2^2}{n_1^2}}$ Substituting for $\sin \theta_r$ in eqn (1) $NA = n_1 \sqrt{\frac{n_1^2 - n_2^2}{n_1^2}}$ Numerical Aperture, $NA = \sqrt{n_1^2 - n_2^2}$ $\sin \theta_c = \frac{n_2}{n_1} \dots \dots \dots (2)$ But $\sin \theta_c = \cos (90 - \theta_c)$ $\theta_c + \theta_r = 90^\circ$ $\sin \theta_c = \cos \theta_r$ From (2) $\cos \theta_r = \frac{n_2}{n_1}$ $\cos^2 \theta_r + \sin^2 \theta_r = 1$ $\sin^2 \theta_r = 1 - \cos^2 \theta_r$ $\sin^2 \theta_r = 1 - \frac{n_2^2}{n_1^2}$ $\sin^2 \theta_r = \frac{n_1^2 - n_2^2}{n_1^2}$
	b	The numerical aperture of an optic fibre is 0.5075 and the refractive index of the cladding is 1.475. Calculate the refractive index of the core, acceptance angle, and the critical angle for total internal reflection.
	ans	$\text{Sol. } n_2 = 1.475$ $NA = 0.5075$ $\sqrt{n_1^2 - n_2^2} = 0.5075$ $\sqrt{n_1^2 - 1.475^2} = 0.5075$ $n_1 = 1.5599$ $\theta_c = \sin^{-1} \left(\frac{n_2}{n_1} \right) = \sin^{-1} \left(\frac{1.475}{1.5599} \right) = 71^\circ$ $\theta_a = \sin^{-1}(NA) = \sin^{-1}(0.5075) = 30^\circ 30'$

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